say 2 to 40 milliseconds, and the FFT used to find the frequency spectrum of each segment. These spectra are placed side-by-side, and converted into a grayscale image (low amplitude becomes light, and high amplitude becomes dark). This provides a graphical way of observing how the frequency content of speech changes with time. The segment length is chosen as a tradeoff between *frequency resolution* (favored by longer segments) and *time resolution* (favored by shorter segments).

As demonstrated by the *a* in *rain*, voiced sounds have a periodic time domain waveform, shown in (a), and a frequency spectrum that is a series of regularly spaced harmonics, shown in (b). In comparison, the *s* in *storm*, shows that fricatives have a noisy time domain signal, as in (c), and a noisy spectrum, displayed in (d). These spectra also show the shaping by the format frequencies for both sounds. Also notice that the time-frequency display of the word *rain* looks similar both times it is spoken.

Over a short period, say 25 milliseconds, a speech signal can be approximated by specifying three parameters: (1) the selection of either a periodic or random noise excitation, (2) the frequency of the periodic wave (if used), and (3) the coefficients of the digital filter used to mimic the vocal tract response. Continuous speech can then be synthesized by continually updating these three parameters about 40 times a second. This approach was responsible for one the early commercial successes of DSP: the *Speak & Spell*, a widely marketed electronic learning aid for children. The sound quality of this type of speech synthesis is poor, sounding very mechanical and not quite human. However, it requires a very low data rate, typically only a few kbits/sec.

This is also the basis for the **linear predictive coding** (**LPC**) method of speech compression. Digitally recorded human speech is broken into short segments, and each is characterized according to the three parameters of the model. This typically requires about a dozen bytes per segment, or 2 to 6 kbytes/sec. The segment information is transmitted or stored as needed, and then reconstructed with the speech synthesizer.

Speech recognition algorithms take this a step further by trying to recognize patterns in the extracted parameters. This typically involves comparing the segment information with templates of previously stored sounds, in an attempt to identify the spoken words. The problem is, this method does not work very well. It is useful for some applications, but is far below the capabilities of human listeners. To understand why speech recognition is so difficult for computers, imagine someone unexpectedly speaking the following sentence:

Larger run medical buy dogs fortunate almost when.

Of course, you will not understand the meaning of this sentence, because it has none. More important, you will probably not even understand all of the individual words that were spoken. This is basic to the way that humans